

ASHIRWAD SINHA

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EDUCATION

Master of Computer Applications 2024–2026
Thapar Institute of Engineering & Technology

Bachelor of Computer Applications 2020–2023
Patliputra University, Patna

ACHIEVEMENTS

- **1st Runner-Up (Top 2 of 250+ teams)** — NSSC 2025, IIT Kharagpur (Data Analytics) Dec 2025
- **National Finalist (Top 10)** — SOLIDWORKS AI Hackathon, IIT Madras (YOLOv8, CV) Jan 2026
- **National Finalist** — Serve Smart Hackathon (Jagriti'26), IIT BHU (YOLOv8, OpenCV) Feb 2026
- **National Finalist** — HackWise 2026 (IRIS), IIM Indore Jan 2026

SKILLS

Languages: Python, C/C++, SQL, HTML/CSS.

AI/ML: PyTorch, Scikit-learn, Machine Learning, Deep Learning (CNN, RNN, LSTM, GRU), NLP.

Computer Vision: YOLOv8, OpenCV.

Web & Backend: Flask, Streamlit, REST APIs.

Data & Visualization: Pandas, NumPy, MySQL, Matplotlib, Seaborn.

MLOps: Docker, MLflow, DVC.

Tools & Cloud: Git, GitHub, Jupyter, Google Colab, VS Code, Hugging Face Spaces, Gradio, AWS.

Soft Skills: Analytical thinking, problem solving, teamwork, leadership, communication.

PROJECTS

HPCL Intelligent Cost Database |  [GitHub](#) Jan 2026

- Built a full-stack AI-powered procurement intelligence platform using React, Flask, Sentence Transformers, and REST APIs for automated item standardization and cost analytics.
- Engineered NLP pipelines for canonical item mapping, achieving a 94.4% average confidence score, and built statistical models to flag critical cost outliers for supplier negotiation.

S.E.N.T.I.N.E.L. – (Serve Smart Hackathon, IIT BHU) |  [GitHub](#) |  [Live Demo](#) Feb 2026

- Engineered a YOLOv8-Medium detection pipeline at 1024px resolution with a custom cyclic checkpointing (“relay”) training strategy, achieving 0.715 mAP@50 while keeping the deployed model under 50MB for fast CPU inference.
- Designed an automated scale-calibration algorithm to convert detected object dimensions into real-world measurements, and built a real-time OpenCV visualization dashboard with Test-Time Augmentation, deployed live on Hugging Face Spaces.

LHC Jet Particle Analysis |  [GitHub](#) |  [Live Demo](#) Dec 2025

- Built a multimodal ML pipeline for the HLS4ML dataset, securing **1st Runner-Up at NSSC 2025, IIT Kharagpur**.
- Fine-tuned ResNet-18 for jet image classification (0.82 F1-Score) and deployed a CNN Autoencoder trained exclusively on normal data to flag anomalies via Hugging Face.

CERTIFICATIONS

- [PyTorch for Deep Learning Bootcamp](#) — Udemy (2025)
- [Introduction to Applied Data Science with Python](#) — Simplilearn SkillUp (2025)